

Lock-Based Protocols & Multiple Granularity -- DBMS

Definition: Lock-Based Protocols control concurrent access to data items by requiring transactions to acquire Locks (Shared or Exclusive) before reading or writing data.

1. Detailed Technical Explanation

1. Lock Modes:

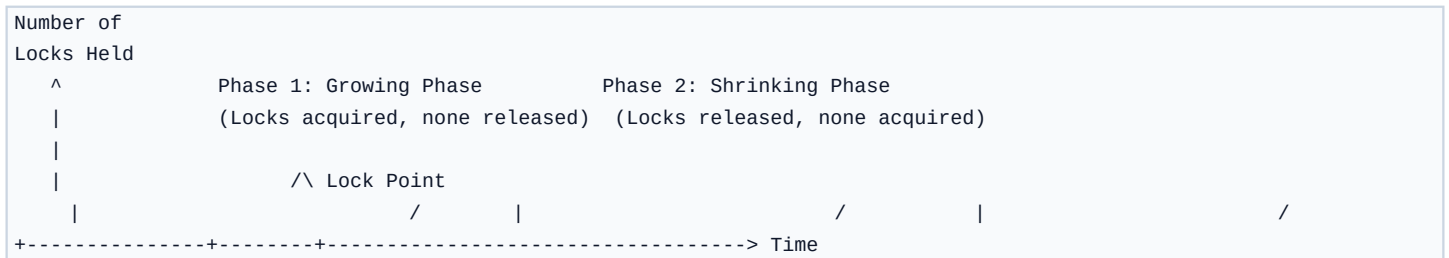
1. Shared Lock (S-Lock): Granted for READ operations. Multiple transactions can hold S-locks simultaneously on the same item.

Lock Compatibility Matrix:

Shared (S)	**Compatible (True)**	Incompatible (False)
Exclusive (X)	Incompatible (False)	Incompatible (False)

2. Two-Phase Locking Protocol (2PL)

2PL guarantees conflict serializability by dividing a transaction's lock lifecycle into two distinct phases:

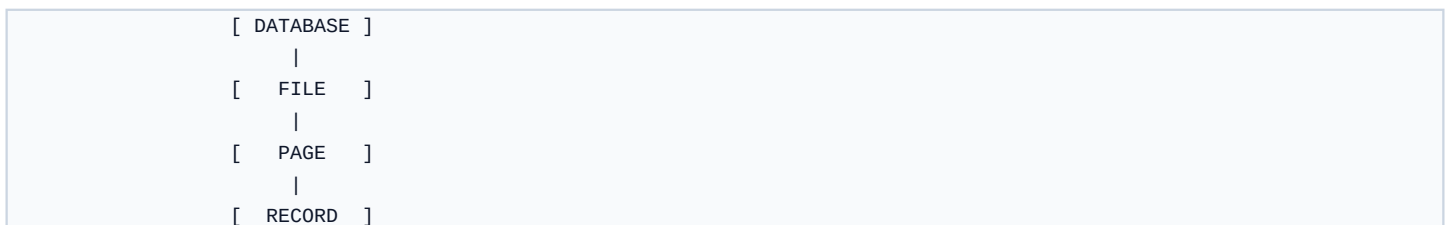


2PL Variants:

1. Basic 2PL: Transaction acquires locks as needed (Growing), releases locks (Shrinking). Can suffer from Cascading Aborts.

3. Multiple Granularity Locking (MGL)

MGL allows data items of various sizes (Database -> File -> Page -> Record) to be locked in a tree hierarchy.



Intent Locks:

Before locking a fine-grained node (e.g., Record), a transaction must acquire an Intent Lock on its ancestor nodes:

4. Core Concepts & Memory Keywords

- * Lock Point: Point in time where a transaction acquires its final lock in 2PL.

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5. Must-Write Points for Exams

- * Basic 2PL guarantees conflict serializability, but does NOT prevent deadlocks.

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6. Quick Recall Flow

Growing Phase (Acquire Locks) -> Lock Point -> Shrinking Phase (Release Locks) -> Hold X-Locks to Commit (Strict 2PL)